CABINET: Content Relevance-Based Noise Reduction for Table Question Answering

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Problem

In table question answering, only a handful of cells hold the answer to a given question; the rest act as **noise**. LLMs are highly susceptible to this irrelevant content and give sub-optimal answers, and the problem **worsens as tables grow larger**.

> *Most of a table is irrelevant to any one question, yet the LLM reads all of it, and gets distracted.*

Motivation

A natural fix is to shrink the table first, but hard decomposition is unforgiving: pick the wrong sub-table and useful cells are lost for good.

DATER: HARD DECOMPOSE

Extracts a sub-table; a wrong pick discards useful cells and locks in the error.

vs.

CABINET: SOFT WEIGHT

Weighs relevant cells higher, removes nothing, keeps the whole table.

Soft weighting de-emphasises noise while keeping every cell available.

Dataset / Benchmark

Evaluated on three challenging table-QA benchmarks; WikiTQ is among the most complex, requiring compositional multi-step reasoning over tables.

WikiTQ FeTaQA WikiSQL

METRICS

WikiTQ & WikiSQL scored by exact-match accuracy; FeTaQA by Sacre-BLEU for long, free-form answers.

The ~300 released parsing statements bootstrap the weakly-supervised cell highlighter.

Method

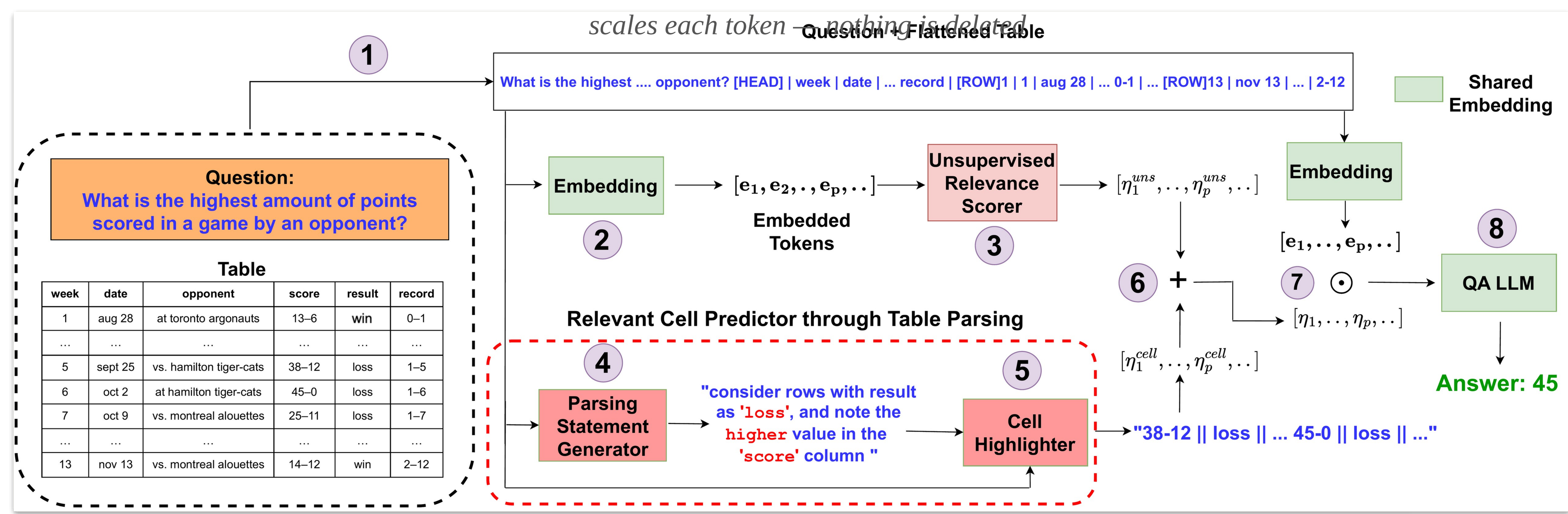
- **Linearize & embed** the table together with the question through the QA LLM's shared embedding.
- **Unsupervised Relevance Scorer** (transformer encoder, variational inference with clustering + separation + sparsification losses) scores every table token.
- **Parsing-statement module** (Flan-T5-XL, ~300 annotations) highlights relevant cells; the two scores are fused and multiply token embeddings before decoding.

$$\mathcal{L} = \mathcal{L}_{CE} + \lambda_{clu}\mathcal{L}_{clu} + \lambda_{sep}\mathcal{L}_{sep} + \lambda_{sparse}\mathcal{L}_{sparse}$$

$$\eta_p = \lambda_{uns}\,\eta_p^{uns} + \lambda_{cell}\,\eta_p^{cell}$$

fused weight *softly*

scales each token — nothing is deleted



CABINET architecture: the unsupervised relevance score and the parsing-statement cell score are fused (steps 6–7) to weigh the table content passed to the QA LLM.

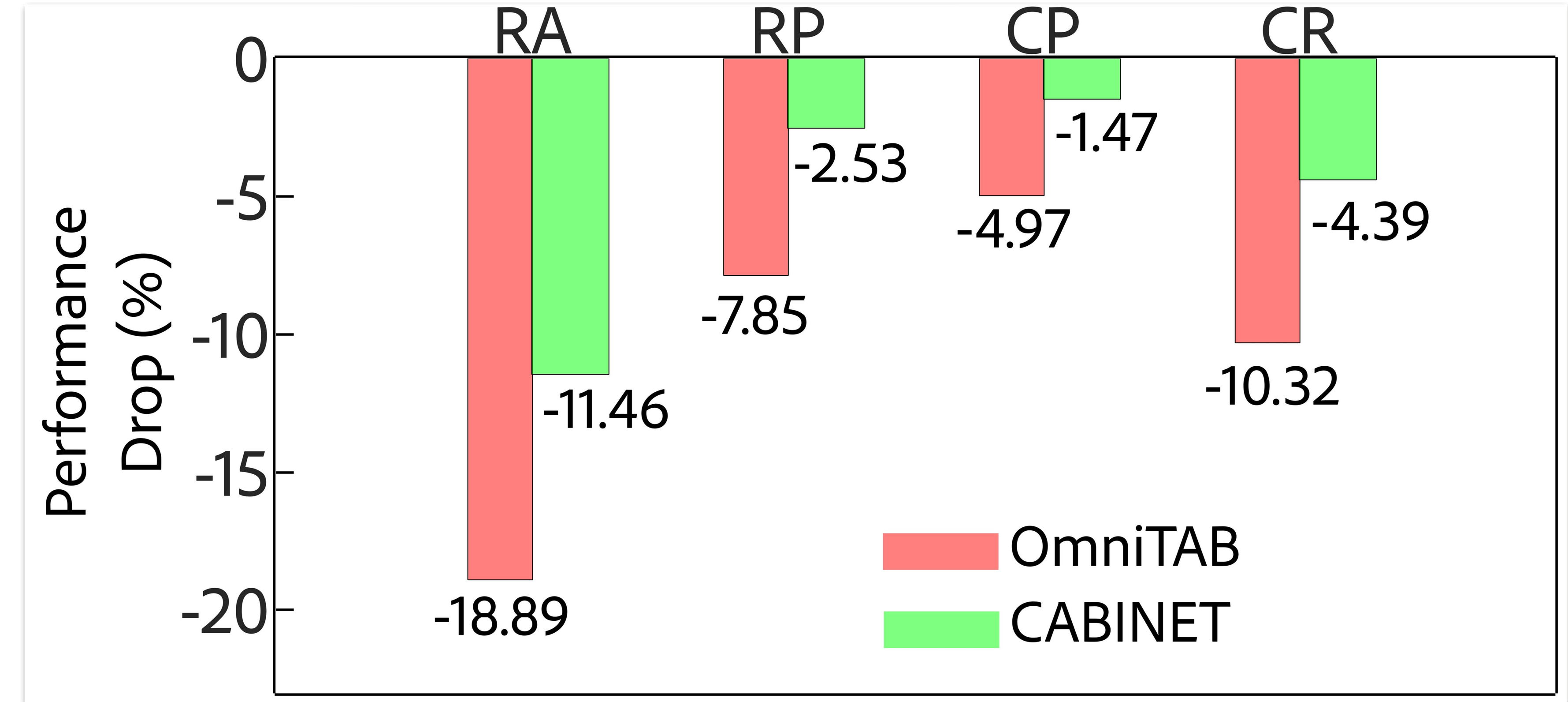
Ablation Study

- **URS losses work only together:** clustering + centroid-separation + sparsification lift WikiTQ 60.8 → 65.6; any subset barely helps.
- **Fuse both signals:** λ_{uns}=0.7, λ_{cell}=0.3 gives 69.1; the cell-based signal alone collapses to 37.6.

The unsupervised scorer is the primary driver, and the weakly-supervised parsing-statement module is a complementary aid.

Key Results

METHOD (WIKITQ)	ACC.	Δ
OmniTab	62.7	−6.4
DATER (GPT-3, in-context)	65.9	−3.2
CABINET (Ours)	**69.1**	—



Performance drop under table perturbations: CABINET (green) is far more robust to noise than OmniTab (red).

Headline Numbers

69.1%

WikiTQ accuracy · new SoTA

+6.4% over OmniTab

40.5 FeTaQA Sacre-BLEU

89.5% WikiSQL accuracy

560M model parameters

Takeaway

Softly weighting table content by learned relevance, instead of hard-decomposing the table, lets an LLM focus on the cells that matter while keeping full access to the data.

Don't cut the table down — weight it. Soft relevance sets new state of the art on three table-QA benchmarks with a compact 560M model.